Hoa Huynh

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Google Scholar: scholar.google.com/citations?user=abuFJTkAAAAJ **Portfolio:** huynhthaihoa.github.io
GitHub: github.com/huynhthaihoa **LeetCode:** leetcode.com/u/huynhthaihoa

ABOUT

Computer Vision Engineer with 7+ years in the field, including 5+ in industry developing and deploying real-time deep learning systems on edge devices for automotive safety and surveillance. Experienced in 2D/3D perception, inference optimization, and multi-sensor integration. Interested in visual SLAM, 3D vision, and robotics.

SKILLS

Programming Languages: C, C++, Python, C#, MATLAB

Perception & Deep Learning: Open3D, PCL, OpenCV, PyTorch, TensorFlow, Darknet

Edge AI Deployment: TI Edge AI/TIOVX (SK-TDA4VM, TDA4AEN), NVIDIA Jetson Nano/Xavier NX (TensorRT, DeepStream), Samsung Exynos V920, ONNX Runtime, TFLite

Sensors & Calibration: RGB/IR pinhole/fisheye, FLIR thermal, Intel RealSense, LUCID Helios2 ToF, Ouster LiDAR

MLOps & Tooling: Git, SVN, Docker, ROS, CMake, TensorBoard, DVC, W&B, MLflow, Jira, Confluence

Databases: Microsoft SQL Server, MySQL, SQLite, PostgreSQL, MongoDB

Additional Languages & Frameworks: Node.js, React, JavaScript, Java, LaTeX, WPF, Qt, FFmpeg

Spoken Languages: English (professional), Korean (elementary), Vietnamese (native)

EXPERIENCE

DeltaX Seoul, South Korea
Development Team Lead & AI Researcher *June 2022 – July 2026*

- Led an R&D team from December 2023, driving the architecture, technical roadmap, and end-to-end delivery of an In-Cabin Monitoring System (ICMS) that aligned with Euro NCAP standards, while coordinating module development, milestone tracking, and client demonstrations across cross-functional teams.
- Architected a real-time edge AI pipeline using a single RGB/IR fisheye camera, C/C++ inference on an SoC (System-on-Chip), and a Node.js/React web visualization UI.
- Integrated 8 deep learning models (object detection, 2D/3D body keypoint estimation, facial landmark detection, gaze estimation, and image classification), achieving an average of 14 FPS on an 8-TOPS edge platform and 9 FPS on a 4-TOPS edge platform.
- Delivered 4 Proof-of-Concept (PoC) projects for 2 Korean automotive vendors within 3–9-month timelines: 42dot (Phase 1: January – June 2024; Phase 2: August – October 2024), Pyeong Hwa Automotive (Phase 1: May 2024 – January 2025; Phase 2: July – October 2025).
- Customized PyTorch models and optimized Python/C++ integration pipelines for SoC targets, deploying on TI SK-TDA4VM/TDA4AEN using Edge AI + TIOVX frameworks, and on Samsung Exynos V920 using an ONNX Runtime-based framework.
- Developed the 3D perception stack for worker localization, 3D display, and ADAS applications (Advanced Driver-Assistance Systems): monocular depth estimation (with pseudo-LiDAR conversion) trained under LiDAR/RGB-D supervision, semantic segmentation, visual SLAM, and real-time multi-camera stitching from fisheye and pinhole inputs.
- Designed a slicing-based preprocessing scheme inspired by SAHI ("Slicing Aided Hyper Inference and Fine-tuning for Small Object Detection") for automotive body panel surface inspection to improve defect detection recall while preserving real-time inference performance.
- Built the team's multi-sensor calibration, data collection, and analysis tooling (ROS, MATLAB, Python), and assembled real-world ADAS/in-cabin datasets across diverse subjects and lighting conditions.

Smartdeep Seoul, South Korea
Research Engineer *May 2021 – May 2022*

- Built a 16-channel CCTV surveillance system in C++ (OpenCV, FFmpeg) with a C# WPF front end, aligned with KISA Intelligent CCTV certification requirements for moving- and abnormal-object detection, recording, and event search.
- Owned dataset curation, training, and validation pipelines for Darknet YOLOv4 detection models.

Ubay Solution Seoul, South Korea
AI Research Intern *December 2020 – April 2021*

- Developed an OCR system for Busan Port to read truck license plates and container codes, benchmarking SSD, YOLOv4, and Faster R-CNN on a custom dataset expanded via augmentation and web crawling.
- Built a DeepStream + TensorRT pipeline on NVIDIA Jetson Nano/Xavier NX for real-time inference.

ANDA Lab – Soongsil University Seoul, South Korea
Research Assistant *September 2018 – August 2020*

- Diagnosed failure modes in a taillight-based vehicle localization system via experimental analysis in Python, OpenCV, and MATLAB, then designed the algorithms to address them: overlapping-LED detection for vehicular visible light communication (VLC), LED taillight detection for optical camera communication (OCC), and stereo-vision taillight matching and pairing for vehicle-to-vehicle positioning.
- First-authored 3 peer-reviewed papers on camera-based nighttime vehicle positioning in IEEE Access (SJR Q1), MDPI Applied Sciences (SJR Q2), and J-KICS (SJR Q4).

Global Cybersoft JSC Ho Chi Minh City, Vietnam
Software Engineer Trainee *March 2017 – August 2017*

- Built C++ plugins for Japanese clients: MFC plugins for SCREEN Holdings' industrial printer system, and Adobe Illustrator SDK plugins (C++, JavaScript) for an advertising agency.

EDUCATION

Soongsil University Seoul, South Korea
Master of Engineering, Information and Communication Technology *September 2018 – August 2020*

- **GPA:** 4.01/4.5
- **Thesis:** LED Taillight Matching Algorithm for Nighttime Vehicle Positioning *(written in English)*
- **Supervisor:** Professor Myungsik Yoo

University of Information Technology – Vietnam National University Ho Chi Minh City, Vietnam
Bachelor of Engineering, Software Engineering *August 2013 – June 2018*

- **GPA:** 8.6/10 (Salutatorian Honors)
- **Thesis:** Blood Vessel Segmentation on DICOM Images *(written in Vietnamese)*
- **Supervisor:** Thi-Vuong Pham, M.Sc.

PUBLICATIONS

*(**J**: International Journal, **W**: International Workshop, **K**: Korean Journal/Conference)*

K2. Uddin S M Nadim et al. (including **Huynh Thai Hoa**), "A Railway Framework for Person (Worker) Localization based on Depth Estimation and GPS", *The Korean Institute of Defense Technology Fall Conference* (2024). Reference

W1. Jaime Spencer et al. (including **Huynh Thai Hoa**), "The Second Monocular Depth Estimation Challenge", *The IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops* (2023). Reference

J2. **Thai-Hoa Huynh**, Myungsik Yoo, "A Taillight Matching and Pairing Algorithm for Stereo-Vision-Based Nighttime Vehicle-to-Vehicle Positioning", *MDPI Applied Sciences* (2020). Reference

K1. **Thai-Hoa Huynh**, Myungsik Yoo, "Nighttime LED Taillight Detection Method with Optical Camera Communication System", *The Journal of Korean Institute of Communications and Information Sciences* (2020). Reference

J1. **Thai-Hoa Huynh**, Tuan-Anh Pham, Myungsik Yoo, "Detection Algorithm for Overlapping LEDs in Vehicular Visible Light Communication System", *IEEE Access* (2019). Reference

SELECTED PROJECTS

State Estimation Simulations: Python simulations of robot state estimation on SO(3)/SE(3): recursive filters (EKF, invariant EKF, UKF) and batch/incremental optimization (factor graphs, iSAM, bundle adjustment), plus LiDAR ICP and legged-robot filtering; tested with pytest and Monte Carlo NEES consistency checks. GitHub

Monocular Metric Depth Estimation: LeReS-based PyTorch pipeline with pluggable encoders (e.g., DINOv2, ConvNeXt) and decoders, benchmarked on NYUv2 and KITTI datasets, with support for TI TIDL quantization-aware training and ONNX/TFLite export for edge deployment. GitHub

Notch-Based Point Cloud Registration: C++ point-cloud registration aligning 3D scans to a reference model via notch-keypoint correspondence matching, for industrial inspection and quality verification. GitHub

LiDAR-Based Ego Lane Detection: Python signal-processing pipeline (ground extraction, intensity analysis, polynomial fitting) for interpretable ego-lane detection directly from LiDAR point clouds. GitHub

Camera Calibration Tool: Python/OpenCV toolkit for estimating camera intrinsics and removing lens distortion across fisheye, pinhole, and Double Sphere models on RGB and thermal sensors, with checkerboard detection, reprojection-error reporting, and stereo calibration. GitHub

(Full project list available on portfolio: huynhthaihoa.github.io/#projects)

SELECTED CERTIFICATIONS

Coursera: Introduction to Machine Learning in Production

Hugging Face: Fine-tuning Language Models, AI Agents Fundamentals, Fundamentals of Group Relative Policy Optimization

(Full certification list available on portfolio: huynhthaihoa.github.io/#certifications)